

Question	Marking guidance	Mark	Comments
08.1	± 0.05 for all values of V_s	1	
08.2	$0.8 (\times 10^{-20})$ error bars added to every point Best fit straight line drawn with a ruler and going through every error bar	1 1	1.6 to 3.2; 7.8 to 9.4; 12.0 to 13.6; 19.2 to 20.8 ($\times 10^{-20}$) Candidates who do not draw error bars can gain the final mark for a good best fit line drawn with the existing points
08.3	Correct data from their graph 6.2×10^{-34} to 7.1×10^{-34} (J s) ecf from best fit line	1 1	Maximum of 1 mark if no working visible on graph
08.4	Their intercept on the f axis correct	1	
08.5	Draws 2 extreme lines within all error bars gives the uncertainty as half the range of intercepts	1 1	
08.6	More sets of values OR Repeats <u>and</u> averages OR Greater precision in V OR Greater range of f	1	

Question	Marking guidance	Mark	Comments
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18	D
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Question	Marking guidance	Mark	Comments
06.1	(free) electrons collide with the (mercury) atoms (and transfer energy) ✓	1	Accept atoms collide with electrons Ignore the idea that atoms collide with each other
06.2	mercury atoms de-excite or emit photons of UV / 7.2 eV light ✓ (UV) photons are absorbed by atoms (of powder coating) ✓	2	Not just collide Accept photons transfer (all of) their energy to the atoms (of the powder)
06.3	de-excitation occurs via other levels the idea that smaller energy drops result in lower energy/ lower frequency/ longer wavelength / visible light emitted	2	Accept in steps
06.4	From 3 to 2, and downwards with only one arrow drawn	1	
06.5	Conversion of eV to J seen or implied ✓ Use of $E = \frac{hc}{\lambda}$ ✓ $5.18 \times 10^{-6} \text{ m}$ ecf from 06.4 ✓	3	other possible answers $5.36 \times 10^{-7} \text{ m}$ $2.68 \times 10^{-7} \text{ m}$ $2.54 \times 10^{-7} \text{ m}$ $1.78 \times 10^{-7} \text{ m}$ $1.73 \times 10^{-7} \text{ m}$

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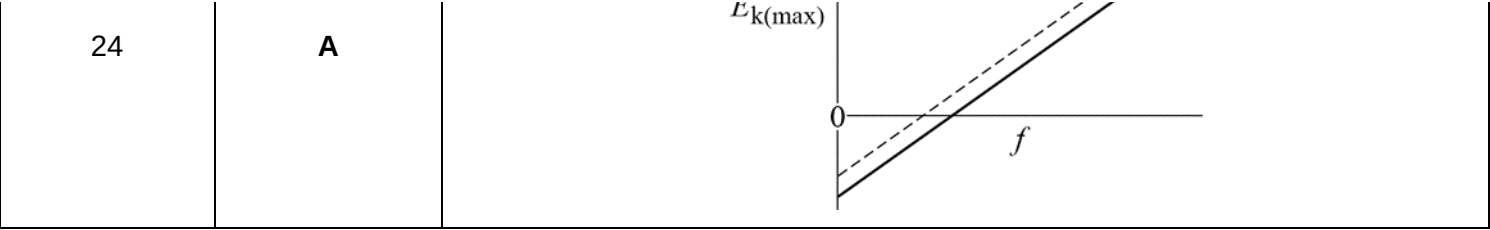
Question	Marking guidance	Additional comments/guidelines	Mark
01.1	The photon (energy) is absorbed OR the atom/electron is excited ✓ (Atom moves to) -1.57 eV level ✓	Evidence for this can come from the diagram i.e. upward arrow to the correct level	2
01.2	6 ✓		1

Question	Marking guidance	Additional comments/guidelines	Mark
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16	B	500 photons m ⁻² s ⁻¹
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22	C	the frequency of the characteristic peaks
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23	C	electromagnetic waves can have particle-like properties.
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Use $Q = It$ ✓ Number of electrons = $3.1(3) \times 10^{15}$ ✓	Charge = $It = 100 \times 10^{-3} \times 5 \times 10^{-3}$ $= 5.0 \times 10^{-4}$ (C) Condone PoT error for MP1	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Total input energy = QV ✓ = $(5.0 \times 10^{-4} \times 80 \times 10^3) = (40 \text{ J})$ output energy = efficiency × input = $0.02 \times$ their energy input OR Heat energy dissipated = $0.98 \times$ their energy input ✓ Heat energy dissipated = 39.2 J ✓	Accept $P = VI$ and $E = Pt$ Ecf for Q from 06.1	3	3 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Max 3 from Idea that more electrons are emitted (and strike anode) (per second) ✓ So more photons are emitted (per second) ✓ Intensity of X-rays goes up ✓ (Max) frequency of X-rays / (max) energy of X-ray photons stays the same ✓		3	3 × AO3
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	hf = energy of (incident) photon ✓ $E_{k(\max)}$ = maximum kinetic energy of (emitted) electron ✓	Do not accept 'max' for 'maximum'	2	2 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Work function ✓ Minimum energy required to remove electron from surface of metal ✓	Condone missing reference to surface of metal but must have reference to minimum or wtte.	2	2 × AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Blue light photon energy > ϕ or red light photon energy < ϕ ✓ $f_b > f_r$ ✓	Alternative: Frequency of blue light > threshold frequency ✓ Frequency of red light < threshold frequency ✓ Allow arguments in terms of wavelength.	2	2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	More photons per unit time / per second✓ Idea that there is a one-to-one relationship between (incident) photon and (emitted) electron✓ so more electrons ejected (per unit time) and current increases ✓		3	3 × AO3
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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25	c	2λ	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
01	uses $E = hf$ or 2.2×10^{-4} (J) seen ✓ value that rounds to 1.4×10^3 (TeV) ✓		2	2 × AO1
Total			2	
Question	Answers	Additional comments/Guidelines	Mark	AO

Question	Answers	Additional comments/Guidelines	Mark	AO
03	idea that a photon transfers, or an electron receives, a discrete amount of energy OR idea that a photon transfers all its energy to an electron ✓ maximum kinetic energy (of electron) equals photon energy minus work function ✓ idea that some electrons have lower KE because: more energy (than work function) is needed to be emitted OR (extra) energy loss occurs during emission ✓	MP1: Allow 'an electron absorbs a photon'. MP2: Allow idea that electrons with max KE are emitted from surface of metal. MP2: Allow use of '$hf - \phi$' if at least one term is defined. MP3: Condone idea that energy to remove electron varies with depth. Reject idea that the work function varies.	3	3 × AO1
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
03	Max 3 from: ✓ ✓ ✓ idea that (atomic) energy levels are quantised (so) transitions between energy levels have discrete values of ΔE idea that wavelengths/frequencies (of spectral lines) correspond to ΔE OR photon energy equals ΔE idea that lines are “characteristic” for a particular element idea that lines are X-rays due to (very) large ΔE	MP1 and MP2 relate to the quantisation of atomic energy levels. MP3 relates emission of radiation/photon to deexcitation process. MP4 relates to why different elements have different spectral lines. e.g. atoms of different elements have unique energy levels MP5: why the lines are X-rays (rather than e.g. visible light)	3	AO1
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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18	A	J	K	J	
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19	C	$1.4 \times 10^{-21} \text{ J}$	
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Question	Answers	Additional comments/Guidelines	Mark	AO
02	Any correct energy conversion from eV to J or vice versa ✓ 1.7 (eV) ✓	Expect to see 2.81 eV OR 1.76×10^{-19} J	2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Any two of: ✓✓ - idea that (free) electrons collide with the atoms (free) electrons transfer energy to the atoms - The atoms go to higher energy levels	Condone throughout for sodium atoms eg 'sodium (orbital) electrons'	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	MP1 Photon energy corresponds to difference between energy levels ✓ Max 2 of: ✓✓ - MP2 Energy levels (within sodium atom) are discrete/fixed/specific/quantised - MP3 Sodium atoms de-excite emitting a photon - MP4 Wavelength is specific/discrete etc since energies are specific and $E = \frac{hc}{\lambda}$	Reject MP2 if candidate implies that photons have energy levels Accept also: idea that the whole spectrum might not be available if limited by the energy/voltage	3	AO2 AO1 AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Read off largest wavelength = 589.6 nm ✓ Use of $E = \frac{hc}{\lambda}$ ✓ 3.37×10^{-19} (J) ✓	award MP2 for any wavelength in the diagram, condone POT must be at least 3sf for MP3	3	AO3
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
01	Makes m the subject of the de Broglie equation OR substitutes for λ and ν into de Broglie equation ✓ $1.9 \times 10^{-28} \text{ (kg)}$ ✓	MP1: condone POT error for λ Calculator value is $1.87925 \times 10^{-28} \text{ kg}$	2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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13	C	$\frac{1}{\text{gradient}}$ $\frac{f \text{ intercept}}{\text{gradient}}$	AO3
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15	A	remove an electron from the atom.	AO1
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19	D	unchanged unchanged doubled	AO3
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	The minimum frequency of light that results in electrons being emitted (from the metal surface) ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	$E_{k(\max)} = eV_{\text{STOP}}$ OR $E = hf$ used at any point ✓ Use of photoelectric effect equation to find ϕ ✓ 5.1×10^{14} (Hz) ✓		3	AO1

Total			4	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	The atoms de-excite and emit photons ✓ corresponding to the energy of each transition ✓ Idea that only specific wavelengths are emitted, showing that only specific transitions are possible ✓		3	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Any three from: ✓✓✓ • use of $E = \frac{hc}{\lambda}$ • converts from eV to J correctly or vice versa at any time • determines 1.9 eV using the wavelength • shows $-1.51 - -3.39 \text{ eV} = 1.9 \text{ eV}$ Concludes level 3 to level 2 ✓		4	3 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	6		1	AO3

Total			8	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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24	B	ultraviolet photons visible photons	AO2
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25	B	increases increases	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
03	Any two from: ✓✓ • use of $\lambda = \frac{h}{mv}$ OR $\lambda = \frac{h}{p}$ • use of $E = \frac{1}{2}mv^2$ • use of $E = eV$ 0.67 (V) ✓	Both mp1 and mp2 can be awarded for the use of a combination of two equations listed on the left-hand side, eg use of one of: • $V = \frac{h^2}{2me\lambda^2}$ • $v = \frac{p^2}{2me}$ Expect to see $v = 4.85 \times 10^5 \text{ m s}^{-1}$ Accept POT error for mp1, mp2	3	AO2
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Idea that the (electrons in the) atoms are in a bound state / energy is required to ionise the atoms / reference to (electrostatic) attraction between the nucleus and the electron ✓ Idea that the ionisation level is defined as zero energy ✓	Allow idea that energy is released when atom moves to a lower energy level Condone that (potential) energy is lower when the electrons are closer to the nucleus Condone electron for atom Condone 'at infinity' for ionisation	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Downward arrow from highest to lowest level		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Any two from: ✓✓ - correct read-offs of the lines for their transition (to half a small square) OR correct read-off of ΔE for their transition eg -2.5 eV, -10.4 eV OR eg $\Delta E = 0.2 \text{ eV}$ - use of $\Delta E = \frac{hc}{\lambda}$ OR $\Delta E = hf$ and $c = f\lambda$ - any correct conversion from eV to J wavelength corresponding to their chosen transition ✓	Eg if correct transition chosen, $1.6 \times 10^{-7} \text{ (m)}$ If shortest transition chosen, $6.2 \times 10^{-6} \text{ (m)}$	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	Any two of: ✓✓ (Free) electrons (emitted by electrode) are accelerated by pd (in the tube) (Free) electrons collide with mercury atoms (Free) Electrons transfer (some of) their (kinetic) energy to the mercury atoms	Do not allow (for any marking point) confusing the free electrons with the mercury atomic electrons Condone electrons in mercury atoms for mercury atoms	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.5	Any three from: ✓✓✓ - mercury atoms de-excite AND emit ultraviolet photons - the (UV) photons are absorbed by the powder coating atoms - idea that powder coating atoms cascade/de-excite in smaller steps/de-excite emitting photons in the visible spectrum or with less energy than UV - idea of lots of visible wavelengths being emitted to combine to form white light	BP1 and BP2 require 'photons' 'powder coating atoms' can be expressed via different phrases	3	AO1
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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22	B	Photoelectrons are emitted.	AO2
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23	A	fast-moving electrons	AO1
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24	A	$f \quad 2P$	AO2
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25	B	The photoelectric effect demonstrates the particle-like behaviour of light.	AO1
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